THEO MESTRE

Software Engineer

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mestretheo@proton.me

MESTRE Theo

Theo-Mestre

TheoMestre

PERSONAL PROFILE

C++ and C# developer with experience in **game engine development** and **real-time 3D** frameworks, and a strong interest in building reliable, high-performance systems. Motivated by technical **problem-solving**, **system architecture**, and solid software foundations across a **range of applications**.

WORK EXPERIENCE

SOFTWARE ENGINEER VR AND SIMULATION

Projet DeMETeRE - Université de Reims

NOVEMBER 2025 - PRESENT

Building on the initial development of **Reims3D**, I now lead the continued evolution of the platform, expanding its scope with **new objectives** and **use cases**. My responsibilities include **extending existing systems**, **improving scalability**, and adapting the project to new **research and visualization** needs. In parallel, I am designing and developing an **industrial robot discovery simulation** aimed at middle and high school students, focusing on **accessibility**, **clarity**, and adapted **interaction** with **real-world industrial concepts** within a 3D environment.

INTERNSHIP - SOFTWARE ENGINEER VR AND SIMULATION

Projet DeMETeRE - Université de Reims

MAY 2025 - NOVEMBER 2025

Developed Reims3D, a 3D **digital twin** of the city of Reims for desktop and **VR**. Owned the full pipeline, from **procedural city generation** using OSM data to **real-time traffic simulation** with **Unreal Engine**'s ZoneGraph and Mass Traffic. Worked in an **R&D** context using **GitLab**, **Trello**, and **Notion**.

UI AND TECHNICAL PROGRAMMER ON COSY SUSHI

Bubble Raccoon Studio

MARCH 2025 - MAY 2025

Implemented **core UI systems** and **data-driven architecture** for a 3D cosy life-simulation and management game in **Unreal Engine**. Worked within a 12-person team on restaurant management and farming systems, using **Notion** for coordination.

ENGINE PROGRAMMER ON SAPPHIRE

SEPTEMBER 2024 - MARCH 2025

Designed and implemented a custom game engine in **C++** and **OpenGL** as a final project. Built core **systems** and **tooling**, using **Premake** for builds and **Git** for version control. The engine is publicly available on GitHub.

UI AND GRAPHIC PROGRAMMER ON TALES OF SCALES

Bubble Raccoon Studio

MARCH - JUNE 2024

Developed UI systems using **Unity UI Toolkit** and contributed to audio ambiance with **Wwise**. Designed a **procedural grass renderer** optimized for large open-world environments using compute shaders. Collaborated within a 10-person team over a 14-week production.

EDUCATION

Bachelor in Video Game Programming

2021 - 2025

Creajeux - France

General Baccalaureate
with a focus in Mathematics and Physics

2018 - 2021

Institution Chaumontaise Oudinot - France

PERSONAL SKILLS

Programming Languages:

Game Engines:

Tools and Frameworks:
Visual Studio, Premake, WPF, Trello, Notion

Version Control:

Audio:

Languages:
French - Native
English - C1

WHAT I'VE SHIPPED

Software
Sapphire : C++ game engine
Tools and Plugins for Unity

Games
Cosy Sushi : Life-Sim
Tales of Scales : Open World RPG
A Steam Story : 2D Adventure Side Scroller